

ML Preprocessing & Feature Selection

Personalized Diet Recommendation

1. Problem Statement

Personalized diet recommendation is treated as a multi-class classification problem. The objective is to clean patient demographic, physical, health, lifestyle and dietary information, select informative predictors, avoid leakage, and prepare an ML-ready dataset for predicting Recommended_Meal_Plan.

2. Dataset Description

The supplied notebook reports 5,000 observations and 30 columns. The target is Recommended_Meal_Plan with four classes: Balanced Diet, High-Protein Diet, Low-Carb Diet and Low-Fat Diet.

3. Preprocessing Techniques

Missing-value audit; numeric median imputation; categorical Unknown/Not Reported treatment; whitespace normalization; exact duplicate detection; domain-plausibility diagnostics; IQR and Z-score outlier detection; conservative outlier retention; standardization; one-hot encoding; and an 80/20 train-test split.

4. Leakage Prevention

Patient_ID is excluded. Recommended_Calories, Recommended_Protein, Recommended_Carbs and Recommended_Fats are treated as potentially target-derived outputs and excluded from predictive candidates.

5. Feature Selection

Variance Threshold, Pearson Correlation, Chi-Square, ANOVA F-Test and Mutual Information are implemented, including from-scratch calculations.

6. Final Selected Features

Weight_kg, BMI, Protein_Intake and Dietary_Habits. The notebook records exactly these four original features.

7. Model Demonstration

The repository adds a reproducible Logistic Regression demonstration. Numeric selected features are median-imputed and standardized; Dietary_Habits is one-hot encoded. The evaluation uses the notebook's 80/20 random split with random_state=42.

8. Result

Accuracy: 24.9%; Macro-F1: 0.2267; Weighted-F1: 0.2266. These are educational ML results on the supplied dataset and are not clinical validation.

9. GUI

src/gui_app.py provides Overview, Data Preview, Feature Selection and Model Demo tabs. The Model Demo accepts Weight_kg, BMI, Protein_Intake and Dietary_Habits and shows the predicted class plus class probabilities.

10. Dataset Source

The uploaded notebook does not document an external URL, Kaggle page, DOI or publisher. Therefore this repository identifies the source as the dataset supplied with the practical assignment/project materials.

Selected Feature Table

Selected Feature	Type	Role
Weight_kg	Numerical	Input
BMI	Numerical	Input
Protein_Intake	Numerical	Input
Dietary_Habits	Categorical	Input